Signal/background separation · mk2 · round 1

```
date         2026-10-25 17:10
data         C:\Users\natha\OneDrive\Desktop\0_mL_ATLAS\0_Sig_BG_09_21_KJ\kj_092126\run3_csv_for_nathan
background   DataHTMissLow_train_data24_cleaned.csv  rows 36,059
signal       HSS_mH125_mS15_ct580_cleaned.csv  rows 570
signal       HSS_mH125_mS55_ct5320_cleaned.csv  rows 423
signal       HSS_mH200_mS50_ct1255_cleaned.csv  rows 1,520
signal       HSS_mH400_mS100_ct1610_cleaned.csv  rows 3,792
signal       HSS_mH600_mS150_ct3310_cleaned.csv  rows 4,490
signal       HSS_mH1000_mS275_ct2399_cleaned.csv  rows 4,188

selections   all  = every event
             ht40_met20 = htmiss_NOSYS >= 40 GeV AND met_met_NOSYS >= 20 GeV, signal and background
region       ('msvtx_inBarrel_NOSYS', 'flag', 'msvtx_inEndcaps_NOSYS')

features common to all 7 files   230
hard / user excluded             15 / 0
non-numeric / constant overall   6 / 20
rank-identical duplicates dropped 91  (in 41 groups; duplicates.txt)
features scored                  98
constant in >= 1 cell            12  (no page in those cells)

metric   two-sample KS D, unweighted events, NaN and -999 removed per feature.
         D_equal weights each mass point 1/N_m; D_pooled concatenates them.
folders  features/NNN_<feature>/, NNN = rank by endcap / ht40_met20 / D_equal

this PDF: cover, selection report, then per selection and region a ranking chart and
top-40 table, then per region the 40 largest drops from all to ht40_met20.
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Selection report

```
cut            htmiss_NOSYS >= 40 GeV  =  40000 file units  (units MeV; signal median 158,509.0 file units)
cut            met_met_NOSYS >= 20 GeV  =  20000 file units  (units MeV; signal median 159,665.4 file units)

per-file edges, in GeV (file units / 1000):
  file          variable                   min     q0.1%       q1%    median   % < cut
  bkg           htmissScalar_NOSYS        0.00      0.00      0.00     26.69
  mH125_mS15    htmissScalar_NOSYS       40.76     41.09     43.13    126.41
  mH125_mS55    htmissScalar_NOSYS       40.05     40.20     42.59    162.50
  mH200_mS50    htmissScalar_NOSYS       40.08     40.14     42.92    127.62
  mH400_mS100   htmissScalar_NOSYS       40.20     41.49     48.66    183.64
  mH600_mS150   htmissScalar_NOSYS       40.03     40.43     44.31    226.81
  mH1000_mS275  htmissScalar_NOSYS       40.49     40.81     47.26    388.57

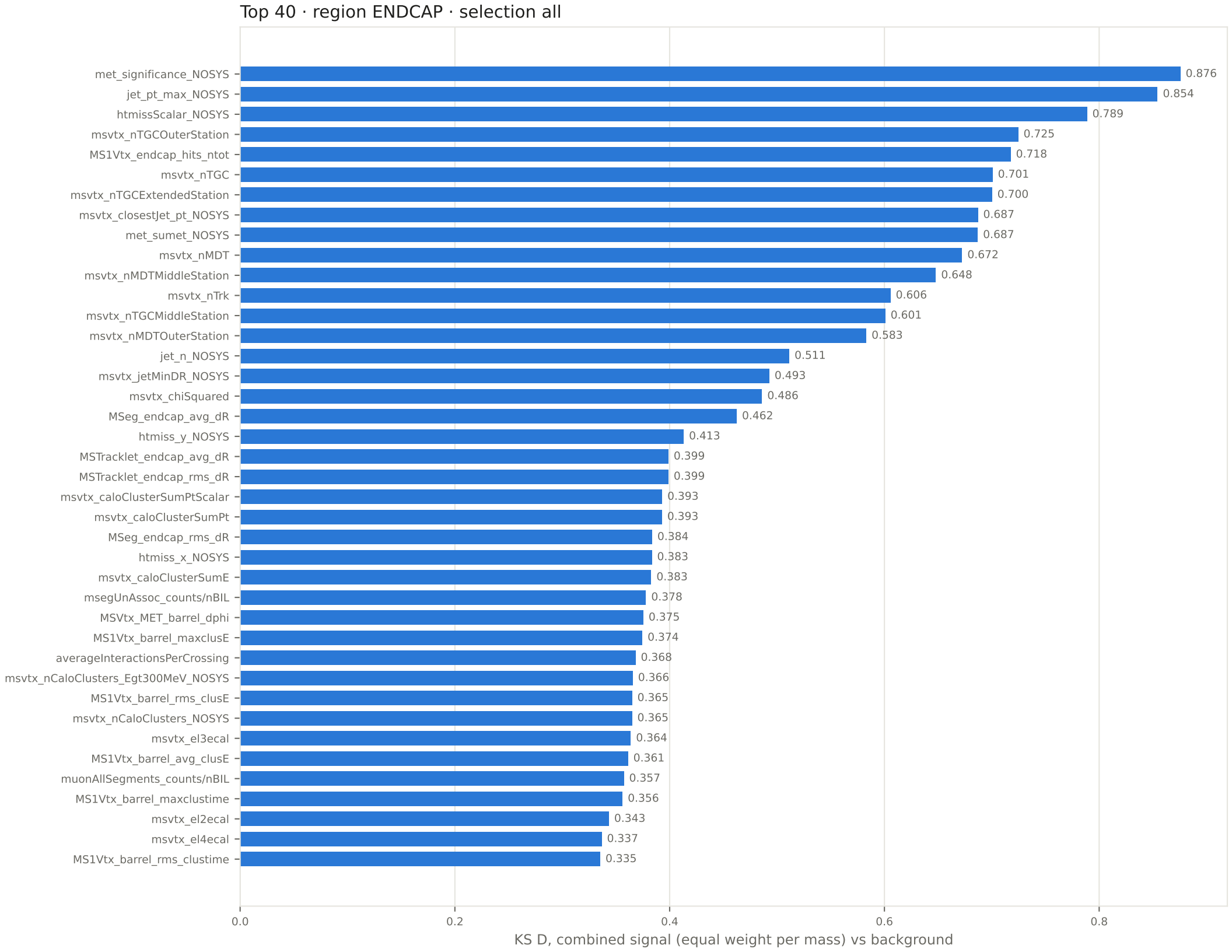
  bkg           htmiss_NOSYS              0.00      0.00      0.00     18.94   100.00%
  mH125_mS15    htmiss_NOSYS             40.36     40.37     41.47     86.20     0.00%
  mH125_mS55    htmiss_NOSYS             40.05     40.10     40.74    125.69     0.00%
  mH200_mS50    htmiss_NOSYS             40.08     40.14     41.05     91.18     0.00%
  mH400_mS100   htmiss_NOSYS             40.20     40.58     44.34    132.91     0.00%
  mH600_mS150   htmiss_NOSYS             40.00     40.22     41.96    164.49     0.00%
  mH1000_mS275  htmiss_NOSYS             40.02     40.77     44.75    279.53     0.00%

  bkg           met_met_NOSYS             0.00      0.42      1.53     18.91    52.46%
  mH125_mS15    met_met_NOSYS            21.55     22.50     24.28     88.31     0.00%
  mH125_mS55    met_met_NOSYS            23.10     23.28     35.81    130.93     0.00%
  mH200_mS50    met_met_NOSYS            20.08     22.03     29.15     92.50     0.00%
  mH400_mS100   met_met_NOSYS            20.07     22.43     35.05    132.38     0.00%
  mH600_mS150   met_met_NOSYS            20.26     21.59     32.90    164.70     0.00%
  mH1000_mS275  met_met_NOSYS            22.32     25.21     39.91    279.95     0.00%

  bkg           jet_pt_max_NOSYS          0.00      0.00      0.00     26.58
  mH125_mS15    jet_pt_max_NOSYS          0.00      0.00     26.46     75.00
  mH125_mS55    jet_pt_max_NOSYS         28.41     29.45     34.30    106.77
  mH200_mS50    jet_pt_max_NOSYS         25.75     27.46     31.08     81.56
  mH400_mS100   jet_pt_max_NOSYS         27.04     28.64     35.26    117.68
  mH600_mS150   jet_pt_max_NOSYS         25.85     28.46     34.42    140.79
  mH1000_mS275  jet_pt_max_NOSYS         26.68     30.42     38.95    231.53

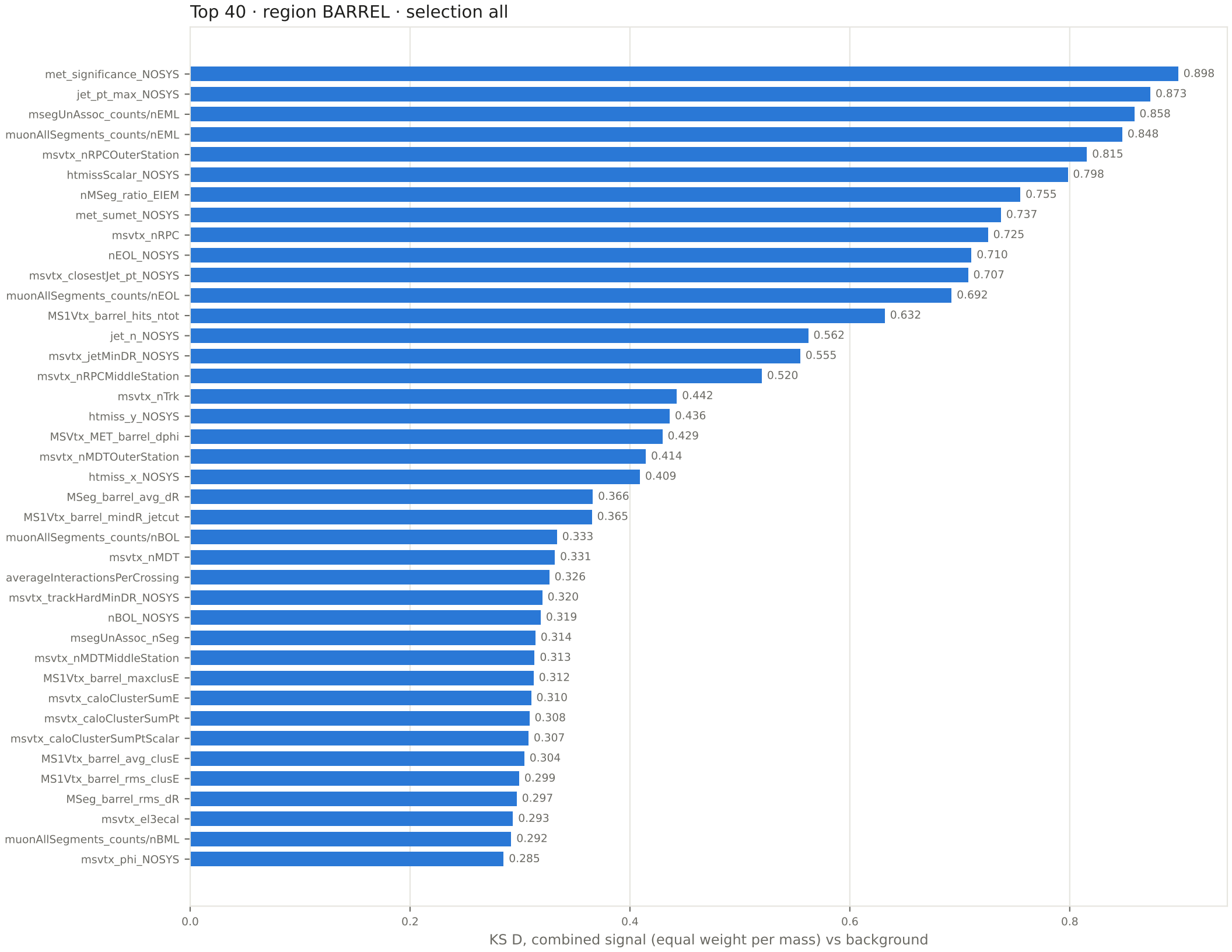
events per file and region, before and after the cut:
  file          region       all ht40_met20    kept    below invalid
  bkg           endcap    34,970        0    0.0%   34,970       0
  bkg           barrel     1,089        0    0.0%    1,089       0
  bkg           all       36,059        0    0.0%   36,059       0
  mH125_mS15    endcap       411      411  100.0%        0       0
  mH125_mS15    barrel       159      159  100.0%        0       0
  mH125_mS15    all          570      570  100.0%        0       0
  mH125_mS55    endcap       281      281  100.0%        0       0
  mH125_mS55    barrel       142      142  100.0%        0       0
  mH125_mS55    all          423      423  100.0%        0       0
  mH200_mS50    endcap     1,071    1,071  100.0%        0       0
  mH200_mS50    barrel       449      449  100.0%        0       0
  mH200_mS50    all        1,520    1,520  100.0%        0       0
  mH400_mS100   endcap     2,648    2,648  100.0%        0       0
  mH400_mS100   barrel     1,144    1,144  100.0%        0       0
  mH400_mS100   all        3,792    3,792  100.0%        0       0
  mH600_mS150   endcap     2,882    2,882  100.0%        0       0
  mH600_mS150   barrel     1,608    1,608  100.0%        0       0
  mH600_mS150   all        4,490    4,490  100.0%        0       0
  mH1000_mS275  endcap     2,703    2,703  100.0%        0       0
  mH1000_mS275  barrel     1,485    1,485  100.0%        0       0
  mH1000_mS275  all        4,188    4,188  100.0%        0       0

all signal files keep >= 99% of events after the cut
```


Top 40 · region ENDCAP · selection all
met_significance_NOSYS 0.876
jet_pt_max_NOSYS 0.854
htmissScalar_NOSYS 0.789
msvtx_nTGCOuterStation 0.725
MS1Vtx_endcap_hits_ntot 0.718
msvtx_nTGC 0.701
msvtx_nTGCExtendedStation 0.700
msvtx_closestJet_pt_NOSYS 0.687
met_sumet_NOSYS 0.687
msvtx_nMDT 0.672
msvtx_nMDTMiddleStation 0.648
msvtx_nTrk 0.606
msvtx_nTGCMiddleStation 0.601
msvtx_nMDTOuterStation 0.583
jet_n_NOSYS 0.511
msvtx_jetMinDR_NOSYS 0.493
msvtx_chiSquared 0.486
MSeg_endcap_avg_dR 0.462
htmiss_y_NOSYS 0.413
MSTracklet_endcap_avg_dR 0.399
MSTracklet_endcap_rms_dR 0.399
msvtx_caloClusterSumPtScalar 0.393
msvtx_caloClusterSumPt 0.393
MSeg_endcap_rms_dR 0.384
htmiss_x_NOSYS 0.383
msvtx_caloClusterSumE 0.383
msegUnAssoc_counts/nBIL 0.378
MSVtx_MET_barrel_dphi 0.375
MS1Vtx_barrel_maxclusE 0.374
averageInteractionsPerCrossing 0.368
msvtx_nCaloClusters_Egt300MeV_NOSYS 0.366
MS1Vtx_barrel_rms_clusE 0.365
msvtx_nCaloClusters_NOSYS 0.365
msvtx_el3ecal 0.364
MS1Vtx_barrel_avg_clusE 0.361
muonAllSegments_counts/nBIL 0.357
MS1Vtx_barrel_maxclustime 0.356
msvtx_el2ecal 0.343
msvtx_el4ecal 0.337
MS1Vtx_barrel_rms_clustime 0.335
0.0
0.2
0.4
0.6
0.8
KS D, combined signal (equal weight per mass) vs background

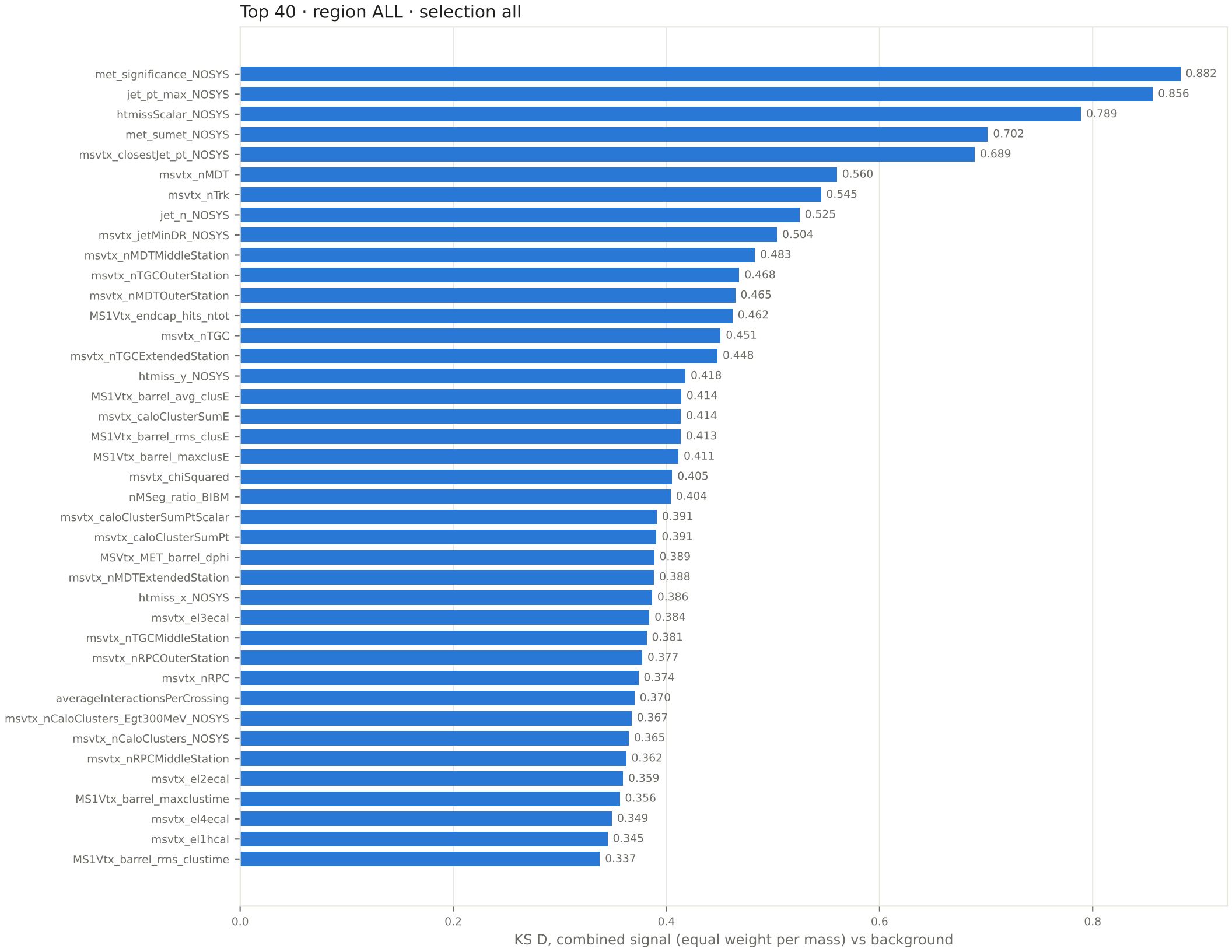
Top 40 table · region ENDCAP · selection all

#	feature	D_equal	D_pooled	D_min	D_max	n_bkg
1	met_significance_NOSYS	0.876	0.896	0.810	0.939	34,970
2	jet_pt_max_NOSYS	0.854	0.874	0.785	0.920	34,970
3	htmissScalar_NOSYS	0.789	0.789	0.789	0.868	34,970
4	msvtx_nTGCOuterStation	0.725	0.785	0.612	0.839	34,970
5	MS1Vtx_endcap_hits_ntot	0.718	0.794	0.553	0.864	34,970
6	msvtx_nTGC	0.701	0.773	0.590	0.840	34,970
7	msvtx_nTGCExtendedStation	0.700	0.766	0.585	0.828	34,970
8	msvtx_closestJet_pt_NOSYS	0.687	0.697	0.647	0.715	34,970
9	met_sumet_NOSYS	0.687	0.741	0.563	0.859	34,970
10	msvtx_nMDT	0.672	0.759	0.491	0.846	34,970
11	msvtx_nMDTMiddleStation	0.648	0.736	0.504	0.827	34,970
12	msvtx_nTrk	0.606	0.607	0.551	0.643	34,970
13	msvtx_nTGCMiddleStation	0.601	0.683	0.471	0.755	34,970
14	msvtx_nMDTOuterStation	0.583	0.636	0.465	0.686	34,970
15	jet_n_NOSYS	0.511	0.556	0.423	0.647	34,970
16	msvtx_jetMinDR_NOSYS	0.493	0.512	0.468	0.538	34,970
17	msvtx_chiSquared	0.486	0.576	0.582	1.000	34,970
18	MSeg_endcap_avg_dR	0.462	0.541	0.235	0.667	34,970
19	htmiss_y_NOSYS	0.413	0.417	0.368	0.449	34,970
20	MSTracklet_endcap_avg_dR	0.399	0.431	0.297	0.502	34,970
21	MSTracklet_endcap_rms_dR	0.399	0.431	0.297	0.502	34,970
22	msvtx_caloClusterSumPtScalar	0.393	0.394	0.389	0.404	34,970
23	msvtx_caloClusterSumPt	0.393	0.393	0.388	0.405	34,970
24	MSeg_endcap_rms_dR	0.384	0.457	0.180	0.594	34,970
25	htmiss_x_NOSYS	0.383	0.398	0.354	0.438	34,970
26	msvtx_caloClusterSumE	0.383	0.386	0.372	0.396	34,970
27	msegUnAssoc_counts/nBIL	0.378	0.346	0.314	0.447	34,970
28	MSVtx_MET_barrel_dphi	0.375	0.357	0.273	0.508	34,970
29	MS1Vtx_barrel_maxclusE	0.374	0.378	0.351	0.386	34,970
30	averageInteractionsPerCrossing	0.368	0.373	0.338	0.391	34,970
31	msvtx_nCaloClusters_Egt300MeV_NOSYS	0.366	0.365	0.355	0.375	34,970
32	MS1Vtx_barrel_rms_clusE	0.365	0.372	0.342	0.384	34,970
33	msvtx_nCaloClusters_NOSYS	0.365	0.364	0.356	0.375	34,970
34	msvtx_el3ecal	0.364	0.366	0.349	0.382	34,970
35	MS1Vtx_barrel_avg_clusE	0.361	0.369	0.336	0.382	34,970
36	muonAllSegments_counts/nBIL	0.357	0.319	0.269	0.426	34,970
37	MS1Vtx_barrel_maxclustime	0.356	0.358	0.345	0.371	34,970
38	msvtx_el2ecal	0.343	0.349	0.336	0.363	34,970
39	msvtx_el4ecal	0.337	0.340	0.293	0.368	34,970
40	MS1Vtx_barrel_rms_clustime	0.335	0.341	0.309	0.356	34,970


Top 40 · region BARREL · selection all
met_significance_NOSYS 0.898
jet_pt_max_NOSYS 0.873
msegUnAssoc_counts/nEML 0.858
muonAllSegments_counts/nEML 0.848
msvtx_nRPCOuterStation 0.815
htmissScalar_NOSYS 0.798
nMSeg_ratio_EIEM 0.755
met_sumet_NOSYS 0.737
msvtx_nRPC 0.725
nEOL_NOSYS 0.710
msvtx_closestJet_pt_NOSYS 0.707
muonAllSegments_counts/nEOL 0.692
MS1Vtx_barrel_hits_ntot 0.632
jet_n_NOSYS 0.562
msvtx_jetMinDR_NOSYS 0.555
msvtx_nRPCMiddleStation 0.520
msvtx_nTrk 0.442
htmiss_y_NOSYS 0.436
MSVtx_MET_barrel_dphi 0.429
msvtx_nMDTOuterStation 0.414
htmiss_x_NOSYS 0.409
MSeg_barrel_avg_dR 0.366
MS1Vtx_barrel_mindR_jetcut 0.365
muonAllSegments_counts/nBOL 0.333
msvtx_nMDT 0.331
averageInteractionsPerCrossing 0.326
msvtx_trackHardMinDR_NOSYS 0.320
nBOL_NOSYS 0.319
msegUnAssoc_nSeg 0.314
msvtx_nMDTMiddleStation 0.313
MS1Vtx_barrel_maxclusE 0.312
msvtx_caloClusterSumE 0.310
msvtx_caloClusterSumPt 0.308
msvtx_caloClusterSumPtScalar 0.307
MS1Vtx_barrel_avg_clusE 0.304
MS1Vtx_barrel_rms_clusE 0.299
MSeg_barrel_rms_dR 0.297
msvtx_el3ecal 0.293
muonAllSegments_counts/nBML 0.292
msvtx_phi_NOSYS 0.285
0.0
0.2
0.4
0.6
0.8
KS D, combined signal (equal weight per mass) vs background

Top 40 table · region BARREL · selection all

#	feature	D_equal	D_pooled	D_min	D_max	n_bkg
1	met_significance_NOSYS	0.898	0.919	0.831	0.962	1,089
2	jet_pt_max_NOSYS	0.873	0.892	0.789	0.916	1,089
3	msegUnAssoc_counts/nEML	0.858	0.843	0.821	0.894	1,089
4	muonAllSegments_counts/nEML	0.848	0.828	0.799	0.901	1,089
5	msvtx_nRPCOuterStation	0.815	0.843	0.740	0.881	1,089
6	htmissScalar_NOSYS	0.798	0.818	0.798	0.872	1,089
7	nMSeg_ratio_EIEM	0.755	0.746	0.727	0.805	1,089
8	met_sumet_NOSYS	0.737	0.783	0.588	0.852	1,089
9	msvtx_nRPC	0.725	0.787	0.647	0.850	1,089
10	nEOL_NOSYS	0.710	0.651	0.580	0.844	1,089
11	msvtx_closestJet_pt_NOSYS	0.707	0.728	0.639	0.740	1,089
12	muonAllSegments_counts/nEOL	0.692	0.632	0.563	0.821	1,089
13	MS1Vtx_barrel_hits_ntot	0.632	0.727	0.405	0.834	1,089
14	jet_n_NOSYS	0.562	0.611	0.458	0.691	1,089
15	msvtx_jetMinDR_NOSYS	0.555	0.583	0.510	0.622	1,089
16	msvtx_nRPCMiddleStation	0.520	0.611	0.314	0.691	1,089
17	msvtx_nTrk	0.442	0.458	0.418	0.475	1,089
18	htmiss_y_NOSYS	0.436	0.437	0.374	0.470	1,089
19	MSVtx_MET_barrel_dphi	0.429	0.407	0.332	0.620	1,089
20	msvtx_nMDTOuterStation	0.414	0.565	0.080	0.710	1,089
21	htmiss_x_NOSYS	0.409	0.421	0.363	0.458	1,089
22	MSeg_barrel_avg_dR	0.366	0.497	0.101	0.645	1,089
23	MS1Vtx_barrel_mindR_jetcut	0.365	0.391	0.307	0.432	1,089
24	muonAllSegments_counts/nBOL	0.333	0.425	0.151	0.517	1,089
25	msvtx_nMDT	0.331	0.500	0.161	0.687	1,089
26	averageInteractionsPerCrossing	0.326	0.324	0.307	0.367	1,089
27	msvtx_trackHardMinDR_NOSYS	0.320	0.346	0.273	0.393	1,089
28	nBOL_NOSYS	0.319	0.408	0.142	0.508	1,089
29	msegUnAssoc_nSeg	0.314	0.175	0.113	0.652	1,089
30	msvtx_nMDTMiddleStation	0.313	0.444	0.196	0.573	1,089
31	MS1Vtx_barrel_maxclusE	0.312	0.314	0.275	0.332	1,089
32	msvtx_caloClusterSumE	0.310	0.312	0.282	0.343	1,089
33	msvtx_caloClusterSumPt	0.308	0.308	0.284	0.346	1,089
34	msvtx_caloClusterSumPtScalar	0.307	0.308	0.284	0.347	1,089
35	MS1Vtx_barrel_avg_clusE	0.304	0.310	0.252	0.333	1,089
36	MS1Vtx_barrel_rms_clusE	0.299	0.309	0.261	0.331	1,089
37	MSeg_barrel_rms_dR	0.297	0.414	0.133	0.584	1,089
38	msvtx_el3ecal	0.293	0.300	0.251	0.331	1,089
39	muonAllSegments_counts/nBML	0.292	0.371	0.097	0.477	1,089
40	msvtx_phi_NOSYS	0.285	0.285	0.279	0.292	1,089


Top 40 · region ALL · selection all
met_significance_NOSYS 0.882
jet_pt_max_NOSYS 0.856
htmissScalar_NOSYS 0.789
met_sumet_NOSYS 0.702
msvtx_closestJet_pt_NOSYS 0.689
msvtx_nMDT 0.560
msvtx_nTrk 0.545
jet_n_NOSYS 0.525
msvtx_jetMinDR_NOSYS 0.504
msvtx_nMDTMiddleStation 0.483
msvtx_nTGCOuterStation 0.468
msvtx_nMDTOuterStation 0.465
MS1Vtx_endcap_hits_ntot 0.462
msvtx_nTGC 0.451
msvtx_nTGCExtendedStation 0.448
htmiss_y_NOSYS 0.418
MS1Vtx_barrel_avg_clusE 0.414
msvtx_caloClusterSumE 0.414
MS1Vtx_barrel_rms_clusE 0.413
MS1Vtx_barrel_maxclusE 0.411
msvtx_chiSquared 0.405
nMSeg_ratio_BIBM 0.404
msvtx_caloClusterSumPtScalar 0.391
msvtx_caloClusterSumPt 0.391
MSVtx_MET_barrel_dphi 0.389
msvtx_nMDTExtendedStation 0.388
htmiss_x_NOSYS 0.386
msvtx_el3ecal 0.384
msvtx_nTGCMiddleStation 0.381
msvtx_nRPCOuterStation 0.377
msvtx_nRPC 0.374
averageInteractionsPerCrossing 0.370
msvtx_nCaloClusters_Egt300MeV_NOSYS 0.367
msvtx_nCaloClusters_NOSYS 0.365
msvtx_nRPCMiddleStation 0.362
msvtx_el2ecal 0.359
MS1Vtx_barrel_maxclustime 0.356
msvtx_el4ecal 0.349
msvtx_el1hcal 0.345
MS1Vtx_barrel_rms_clustime 0.337
0.0
0.2
0.4
0.6
0.8
KS D, combined signal (equal weight per mass) vs background

Top 40 table · region ALL · selection all

#	feature	D_equal	D_pooled	D_min	D_max	n_bkg
1	met_significance_NOSYS	0.882	0.902	0.814	0.938	36,059
2	jet_pt_max_NOSYS	0.856	0.876	0.782	0.913	36,059
3	htmissScalar_NOSYS	0.789	0.792	0.789	0.868	36,059
4	met_sumet_NOSYS	0.702	0.753	0.563	0.856	36,059
5	msvtx_closestJet_pt_NOSYS	0.689	0.703	0.640	0.719	36,059
6	msvtx_nMDT	0.560	0.666	0.366	0.783	36,059
7	msvtx_nTrk	0.545	0.548	0.502	0.569	36,059
8	jet_n_NOSYS	0.525	0.572	0.422	0.659	36,059
9	msvtx_jetMinDR_NOSYS	0.504	0.525	0.480	0.548	36,059
10	msvtx_nMDTMiddleStation	0.483	0.573	0.321	0.670	36,059
11	msvtx_nTGCOuterStation	0.468	0.505	0.371	0.532	36,059
12	msvtx_nMDTOuterStation	0.465	0.543	0.326	0.634	36,059
13	MS1Vtx_endcap_hits_ntot	0.462	0.511	0.335	0.547	36,059
14	msvtx_nTGC	0.451	0.498	0.354	0.527	36,059
15	msvtx_nTGCExtendedStation	0.448	0.490	0.351	0.519	36,059
16	htmiss_y_NOSYS	0.418	0.422	0.364	0.447	36,059
17	MS1Vtx_barrel_avg_clusE	0.414	0.421	0.396	0.429	36,059
18	msvtx_caloClusterSumE	0.414	0.415	0.411	0.420	36,059
19	MS1Vtx_barrel_rms_clusE	0.413	0.420	0.399	0.428	36,059
20	MS1Vtx_barrel_maxclusE	0.411	0.414	0.394	0.419	36,059
21	msvtx_chiSquared	0.405	0.490	0.494	1.000	36,059
22	nMSeg_ratio_BIBM	0.404	0.410	0.381	0.427	36,059
23	msvtx_caloClusterSumPtScalar	0.391	0.392	0.378	0.403	36,059
24	msvtx_caloClusterSumPt	0.391	0.392	0.377	0.403	36,059
25	MSVtx_MET_barrel_dphi	0.389	0.368	0.292	0.536	36,059
26	msvtx_nMDTExtendedStation	0.388	0.358	0.344	0.468	36,059
27	htmiss_x_NOSYS	0.386	0.401	0.356	0.444	36,059
28	msvtx_el3ecal	0.384	0.389	0.368	0.403	36,059
29	msvtx_nTGCMiddleStation	0.381	0.435	0.318	0.471	36,059
30	msvtx_nRPCOuterStation	0.377	0.419	0.306	0.478	36,059
31	msvtx_nRPC	0.374	0.417	0.303	0.476	36,059
32	averageInteractionsPerCrossing	0.370	0.374	0.338	0.396	36,059
33	msvtx_nCaloClusters_Egt300MeV_NOSYS	0.367	0.368	0.357	0.383	36,059
34	msvtx_nCaloClusters_NOSYS	0.365	0.366	0.352	0.383	36,059
35	msvtx_nRPCMiddleStation	0.362	0.401	0.293	0.454	36,059
36	msvtx_el2ecal	0.359	0.366	0.334	0.375	36,059
37	MS1Vtx_barrel_maxclustime	0.356	0.360	0.330	0.372	36,059
38	msvtx_el4ecal	0.349	0.355	0.309	0.379	36,059
39	msvtx_el1hcal	0.345	0.348	0.330	0.357	36,059
40	MS1Vtx_barrel_rms_clustime	0.337	0.344	0.292	0.355	36,059

Top 40 table · region ENDCAP · selection ht40_met20

#	feature	D_equal	D_pooled	D_min	D_max	n_bkg
1	MS1Vtx_barrel_avg_clusE	nan	nan	nan	nan	0
2	MS1Vtx_barrel_avg_clustime	nan	nan	nan	nan	0
3	MS1Vtx_barrel_hits_ntot	nan	nan	nan	nan	0
4	MS1Vtx_barrel_maxclusE	nan	nan	nan	nan	0
5	MS1Vtx_barrel_maxclustime	nan	nan	nan	nan	0
6	MS1Vtx_barrel_mindR_jetcut	nan	nan	nan	nan	0
7	MS1Vtx_barrel_rms_clusE	nan	nan	nan	nan	0
8	MS1Vtx_barrel_rms_clustime	nan	nan	nan	nan	0
9	MS1Vtx_endcap_hits_ntot	nan	nan	nan	nan	0
10	MSTracklet_barrel_avg_dR	nan	nan	nan	nan	0
11	MSTracklet_barrel_rms_dR	nan	nan	nan	nan	0
12	MSTracklet_endcap_avg_dR	nan	nan	nan	nan	0
13	MSTracklet_endcap_rms_dR	nan	nan	nan	nan	0
14	MSVtx_MET_barrel_dphi	nan	nan	nan	nan	0
15	MSeg_barrel_avg_dR	nan	nan	nan	nan	0
16	MSeg_barrel_rms_dR	nan	nan	nan	nan	0
17	MSeg_endcap_avg_dR	nan	nan	nan	nan	0
18	MSeg_endcap_rms_dR	nan	nan	nan	nan	0
19	actualInteractionsPerCrossing	nan	nan	nan	nan	0
20	averageInteractionsPerCrossing	nan	nan	nan	nan	0
21	coreFlags	nan	nan	nan	nan	0
22	htmissScalar_NOSYS	nan	nan	nan	nan	0
23	htmiss_x_NOSYS	nan	nan	nan	nan	0
24	htmiss_y_NOSYS	nan	nan	nan	nan	0
25	jet_n_NOSYS	nan	nan	nan	nan	0
26	jet_pt_max_NOSYS	nan	nan	nan	nan	0
27	met_phi_NOSYS	nan	nan	nan	nan	0
28	met_significance_NOSYS	nan	nan	nan	nan	0
29	met_sumet_NOSYS	nan	nan	nan	nan	0
30	msegNCB_counts/nEIL	nan	nan	nan	nan	0
31	msegUnAssoc_counts/nBIL	nan	nan	nan	nan	0
32	msegUnAssoc_counts/nBML	nan	nan	nan	nan	0
33	msegUnAssoc_counts/nEIL	nan	nan	nan	nan	0
34	msegUnAssoc_counts/nEML	nan	nan	nan	nan	0
35	msegUnAssoc_nSeg	nan	nan	nan	nan	0
36	mseg_counts/nBIL	nan	nan	nan	nan	0
37	mseg_counts/nBML	nan	nan	nan	nan	0
38	mseg_counts/nBOL	nan	nan	nan	nan	0
39	mseg_counts/nEIL	nan	nan	nan	nan	0
40	mseg_counts/nEML	nan	nan	nan	nan	0

Top 40 table · region BARREL · selection ht40_met20

#	feature	D_equal	D_pooled	D_min	D_max	n_bkg
1	MS1Vtx_barrel_avg_clusE	nan	nan	nan	nan	0
2	MS1Vtx_barrel_avg_clustime	nan	nan	nan	nan	0
3	MS1Vtx_barrel_hits_ntot	nan	nan	nan	nan	0
4	MS1Vtx_barrel_maxclusE	nan	nan	nan	nan	0
5	MS1Vtx_barrel_maxclustime	nan	nan	nan	nan	0
6	MS1Vtx_barrel_mindR_jetcut	nan	nan	nan	nan	0
7	MS1Vtx_barrel_rms_clusE	nan	nan	nan	nan	0
8	MS1Vtx_barrel_rms_clustime	nan	nan	nan	nan	0
9	MS1Vtx_endcap_hits_ntot	nan	nan	nan	nan	0
10	MSTracklet_barrel_avg_dR	nan	nan	nan	nan	0
11	MSTracklet_barrel_rms_dR	nan	nan	nan	nan	0
12	MSTracklet_endcap_avg_dR	nan	nan	nan	nan	0
13	MSTracklet_endcap_rms_dR	nan	nan	nan	nan	0
14	MSVtx_MET_barrel_dphi	nan	nan	nan	nan	0
15	MSeg_barrel_avg_dR	nan	nan	nan	nan	0
16	MSeg_barrel_rms_dR	nan	nan	nan	nan	0
17	MSeg_endcap_avg_dR	nan	nan	nan	nan	0
18	MSeg_endcap_rms_dR	nan	nan	nan	nan	0
19	actualInteractionsPerCrossing	nan	nan	nan	nan	0
20	averageInteractionsPerCrossing	nan	nan	nan	nan	0
21	coreFlags	nan	nan	nan	nan	0
22	htmissScalar_NOSYS	nan	nan	nan	nan	0
23	htmiss_x_NOSYS	nan	nan	nan	nan	0
24	htmiss_y_NOSYS	nan	nan	nan	nan	0
25	jet_n_NOSYS	nan	nan	nan	nan	0
26	jet_pt_max_NOSYS	nan	nan	nan	nan	0
27	met_phi_NOSYS	nan	nan	nan	nan	0
28	met_significance_NOSYS	nan	nan	nan	nan	0
29	met_sumet_NOSYS	nan	nan	nan	nan	0
30	msegNCB_counts/nEIL	nan	nan	nan	nan	0
31	msegUnAssoc_counts/nBIL	nan	nan	nan	nan	0
32	msegUnAssoc_counts/nBML	nan	nan	nan	nan	0
33	msegUnAssoc_counts/nEIL	nan	nan	nan	nan	0
34	msegUnAssoc_counts/nEML	nan	nan	nan	nan	0
35	msegUnAssoc_nSeg	nan	nan	nan	nan	0
36	mseg_counts/nBIL	nan	nan	nan	nan	0
37	mseg_counts/nBML	nan	nan	nan	nan	0
38	mseg_counts/nBOL	nan	nan	nan	nan	0
39	mseg_counts/nEIL	nan	nan	nan	nan	0
40	mseg_counts/nEML	nan	nan	nan	nan	0

Top 40 table · region ALL · selection ht40_met20

#	feature	D_equal	D_pooled	D_min	D_max	n_bkg
1	MS1Vtx_barrel_avg_clusE	nan	nan	nan	nan	0
2	MS1Vtx_barrel_avg_clustime	nan	nan	nan	nan	0
3	MS1Vtx_barrel_hits_ntot	nan	nan	nan	nan	0
4	MS1Vtx_barrel_maxclusE	nan	nan	nan	nan	0
5	MS1Vtx_barrel_maxclustime	nan	nan	nan	nan	0
6	MS1Vtx_barrel_mindR_jetcut	nan	nan	nan	nan	0
7	MS1Vtx_barrel_rms_clusE	nan	nan	nan	nan	0
8	MS1Vtx_barrel_rms_clustime	nan	nan	nan	nan	0
9	MS1Vtx_endcap_hits_ntot	nan	nan	nan	nan	0
10	MSTracklet_barrel_avg_dR	nan	nan	nan	nan	0
11	MSTracklet_barrel_rms_dR	nan	nan	nan	nan	0
12	MSTracklet_endcap_avg_dR	nan	nan	nan	nan	0
13	MSTracklet_endcap_rms_dR	nan	nan	nan	nan	0
14	MSVtx_MET_barrel_dphi	nan	nan	nan	nan	0
15	MSeg_barrel_avg_dR	nan	nan	nan	nan	0
16	MSeg_barrel_rms_dR	nan	nan	nan	nan	0
17	MSeg_endcap_avg_dR	nan	nan	nan	nan	0
18	MSeg_endcap_rms_dR	nan	nan	nan	nan	0
19	actualInteractionsPerCrossing	nan	nan	nan	nan	0
20	averageInteractionsPerCrossing	nan	nan	nan	nan	0
21	coreFlags	nan	nan	nan	nan	0
22	htmissScalar_NOSYS	nan	nan	nan	nan	0
23	htmiss_x_NOSYS	nan	nan	nan	nan	0
24	htmiss_y_NOSYS	nan	nan	nan	nan	0
25	jet_n_NOSYS	nan	nan	nan	nan	0
26	jet_pt_max_NOSYS	nan	nan	nan	nan	0
27	met_phi_NOSYS	nan	nan	nan	nan	0
28	met_significance_NOSYS	nan	nan	nan	nan	0
29	met_sumet_NOSYS	nan	nan	nan	nan	0
30	msegNCB_counts/nEIL	nan	nan	nan	nan	0
31	msegUnAssoc_counts/nBIL	nan	nan	nan	nan	0
32	msegUnAssoc_counts/nBML	nan	nan	nan	nan	0
33	msegUnAssoc_counts/nEIL	nan	nan	nan	nan	0
34	msegUnAssoc_counts/nEML	nan	nan	nan	nan	0
35	msegUnAssoc_nSeg	nan	nan	nan	nan	0
36	mseg_counts/nBIL	nan	nan	nan	nan	0
37	mseg_counts/nBML	nan	nan	nan	nan	0
38	mseg_counts/nBOL	nan	nan	nan	nan	0
39	mseg_counts/nEIL	nan	nan	nan	nan	0
40	mseg_counts/nEML	nan	nan	nan	nan	0

Largest drops all -> ht40_met20 · region ENDCAP

feature	D_all	D_ht40_met20	delta	D_split
MS1Vtx_barrel_avg_clusE	0.361	nan	nan	nan
MS1Vtx_barrel_avg_clustime	0.229	nan	nan	nan
MS1Vtx_barrel_hits_ntot	nan	nan	nan	nan
MS1Vtx_barrel_maxclusE	0.374	nan	nan	nan
MS1Vtx_barrel_maxclustime	0.356	nan	nan	nan
MS1Vtx_barrel_mindR_jetcut	0.308	nan	nan	nan
MS1Vtx_barrel_rms_clusE	0.365	nan	nan	nan
MS1Vtx_barrel_rms_clustime	0.335	nan	nan	nan
MS1Vtx_endcap_hits_ntot	0.718	nan	nan	nan
MSTracklet_barrel_avg_dR	nan	nan	nan	nan
MSTracklet_barrel_rms_dR	nan	nan	nan	nan
MSTracklet_endcap_avg_dR	0.399	nan	nan	nan
MSTracklet_endcap_rms_dR	0.399	nan	nan	nan
MSVtx_MET_barrel_dphi	0.375	nan	nan	nan
MSeg_barrel_avg_dR	nan	nan	nan	nan
MSeg_barrel_rms_dR	nan	nan	nan	nan
MSeg_endcap_avg_dR	0.462	nan	nan	nan
MSeg_endcap_rms_dR	0.384	nan	nan	nan
actualInteractionsPerCrossing	0.298	nan	nan	nan
averageInteractionsPerCrossing	0.368	nan	nan	nan
coreFlags	0.007	nan	nan	nan
htmissScalar_NOSYS	0.789	nan	nan	nan
htmiss_x_NOSYS	0.383	nan	nan	nan
htmiss_y_NOSYS	0.413	nan	nan	nan
jet_n_NOSYS	0.511	nan	nan	nan
jet_pt_max_NOSYS	0.854	nan	nan	nan
met_phi_NOSYS	0.013	nan	nan	nan
met_significance_NOSYS	0.876	nan	nan	nan
met_sumet_NOSYS	0.687	nan	nan	nan
msegNCB_counts/nEIL	0.151	nan	nan	nan
msegUnAssoc_counts/nBIL	0.378	nan	nan	nan
msegUnAssoc_counts/nBML	0.069	nan	nan	nan
msegUnAssoc_counts/nEIL	0.115	nan	nan	nan
msegUnAssoc_counts/nEML	0.187	nan	nan	nan
msegUnAssoc_nSeg	0.187	nan	nan	nan
mseg_counts/nBIL	0.041	nan	nan	nan
mseg_counts/nBML	0.053	nan	nan	nan
mseg_counts/nBOL	0.057	nan	nan	nan
mseg_counts/nEIL	0.093	nan	nan	nan
mseg_counts/nEML	0.033	nan	nan	nan

delta = D_ht40_met20 - D_all. Large negative: the separation came from events the cut removes.
D_split = background below vs at/above the cut (shape change across the cut).

Largest drops all -> ht40_met20 · region BARREL

feature	D_all	D_ht40_met20	delta	D_split
MS1Vtx_barrel_avg_clusE	0.304	nan	nan	nan
MS1Vtx_barrel_avg_clustime	0.157	nan	nan	nan
MS1Vtx_barrel_hits_ntot	0.632	nan	nan	nan
MS1Vtx_barrel_maxclusE	0.312	nan	nan	nan
MS1Vtx_barrel_maxclustime	0.274	nan	nan	nan
MS1Vtx_barrel_mindR_jetcut	0.365	nan	nan	nan
MS1Vtx_barrel_rms_clusE	0.299	nan	nan	nan
MS1Vtx_barrel_rms_clustime	0.262	nan	nan	nan
MS1Vtx_endcap_hits_ntot	nan	nan	nan	nan
MSTracklet_barrel_avg_dR	0.097	nan	nan	nan
MSTracklet_barrel_rms_dR	0.097	nan	nan	nan
MSTracklet_endcap_avg_dR	nan	nan	nan	nan
MSTracklet_endcap_rms_dR	nan	nan	nan	nan
MSVtx_MET_barrel_dphi	0.429	nan	nan	nan
MSeg_barrel_avg_dR	0.366	nan	nan	nan
MSeg_barrel_rms_dR	0.297	nan	nan	nan
MSeg_endcap_avg_dR	nan	nan	nan	nan
MSeg_endcap_rms_dR	nan	nan	nan	nan
actualInteractionsPerCrossing	0.237	nan	nan	nan
averageInteractionsPerCrossing	0.326	nan	nan	nan
coreFlags	0.010	nan	nan	nan
htmissScalar_NOSYS	0.798	nan	nan	nan
htmiss_x_NOSYS	0.409	nan	nan	nan
htmiss_y_NOSYS	0.436	nan	nan	nan
jet_n_NOSYS	0.562	nan	nan	nan
jet_pt_max_NOSYS	0.873	nan	nan	nan
met_phi_NOSYS	0.041	nan	nan	nan
met_significance_NOSYS	0.898	nan	nan	nan
met_sumet_NOSYS	0.737	nan	nan	nan
msegNCB_counts/nEIL	0.168	nan	nan	nan
msegUnAssoc_counts/nBIL	0.152	nan	nan	nan
msegUnAssoc_counts/nBML	0.277	nan	nan	nan
msegUnAssoc_counts/nEIL	0.136	nan	nan	nan
msegUnAssoc_counts/nEML	0.858	nan	nan	nan
msegUnAssoc_nSeg	0.314	nan	nan	nan
mseg_counts/nBIL	0.097	nan	nan	nan
mseg_counts/nBML	0.135	nan	nan	nan
mseg_counts/nBOL	0.132	nan	nan	nan
mseg_counts/nEIL	0.170	nan	nan	nan
mseg_counts/nEML	0.090	nan	nan	nan

delta = D_ht40_met20 - D_all. Large negative: the separation came from events the cut removes.
D_split = background below vs at/above the cut (shape change across the cut).

Largest drops all -> ht40_met20 · region ALL

feature	D_all	D_ht40_met20	delta	D_split
MS1Vtx_barrel_avg_clusE	0.414	nan	nan	nan
MS1Vtx_barrel_avg_clustime	0.224	nan	nan	nan
MS1Vtx_barrel_hits_ntot	0.294	nan	nan	nan
MS1Vtx_barrel_maxclusE	0.411	nan	nan	nan
MS1Vtx_barrel_maxclustime	0.356	nan	nan	nan
MS1Vtx_barrel_mindR_jetcut	0.324	nan	nan	nan
MS1Vtx_barrel_rms_clusE	0.413	nan	nan	nan
MS1Vtx_barrel_rms_clustime	0.337	nan	nan	nan
MS1Vtx_endcap_hits_ntot	0.462	nan	nan	nan
MSTracklet_barrel_avg_dR	0.043	nan	nan	nan
MSTracklet_barrel_rms_dR	0.043	nan	nan	nan
MSTracklet_endcap_avg_dR	0.238	nan	nan	nan
MSTracklet_endcap_rms_dR	0.238	nan	nan	nan
MSVtx_MET_barrel_dphi	0.389	nan	nan	nan
MSeg_barrel_avg_dR	0.291	nan	nan	nan
MSeg_barrel_rms_dR	0.291	nan	nan	nan
MSeg_endcap_avg_dR	0.291	nan	nan	nan
MSeg_endcap_rms_dR	0.291	nan	nan	nan
actualInteractionsPerCrossing	0.296	nan	nan	nan
averageInteractionsPerCrossing	0.370	nan	nan	nan
coreFlags	0.007	nan	nan	nan
htmissScalar_NOSYS	0.789	nan	nan	nan
htmiss_x_NOSYS	0.386	nan	nan	nan
htmiss_y_NOSYS	0.418	nan	nan	nan
jet_n_NOSYS	0.525	nan	nan	nan
jet_pt_max_NOSYS	0.856	nan	nan	nan
met_phi_NOSYS	0.006	nan	nan	nan
met_significance_NOSYS	0.882	nan	nan	nan
met_sumet_NOSYS	0.702	nan	nan	nan
msegNCB_counts/nEIL	0.129	nan	nan	nan
msegUnAssoc_counts/nBIL	0.238	nan	nan	nan
msegUnAssoc_counts/nBML	0.312	nan	nan	nan
msegUnAssoc_counts/nEIL	0.105	nan	nan	nan
msegUnAssoc_counts/nEML	0.311	nan	nan	nan
msegUnAssoc_nSeg	0.190	nan	nan	nan
mseg_counts/nBIL	0.155	nan	nan	nan
mseg_counts/nBML	0.242	nan	nan	nan
mseg_counts/nBOL	0.252	nan	nan	nan
mseg_counts/nEIL	0.181	nan	nan	nan
mseg_counts/nEML	0.156	nan	nan	nan

delta = D_ht40_met20 - D_all. Large negative: the separation came from events the cut removes.
D_split = background below vs at/above the cut (shape change across the cut).